

Cloud Computing (CS-4075)

Course Instructor(s)

Dr. Basharat Hussain

Section(s)

8A, 8B

FINAL Exam

Total Time (Hrs): 3

Total Marks: 95

Total Questions: 6

Date: 11-June-2026

Roll No

Course Section

Student Signature

YOU MUST READ THESE INSTRUCTIONS.

Instructions:

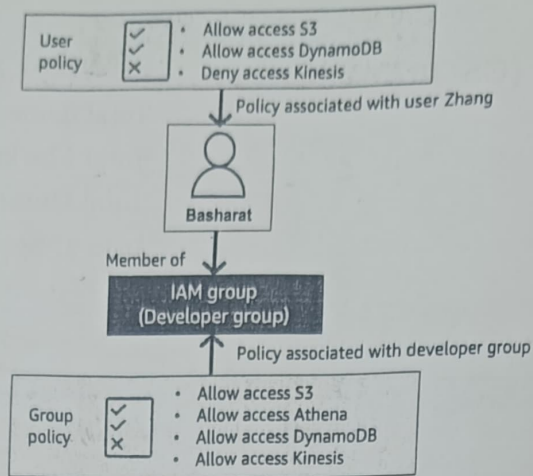
AWS

- DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.
- Attempt ALL questions on the provided Question sheet. Read each question carefully before attempting.
- You have a total of (20) pages and (6) Questions. Please make sure no page is missed in printing.
- Attempt all the questions in given order, including all its sub-parts. [Violation: -5 marks]
- Attempt each new question from the next page. [Violation: -5 marks]
- Calculator sharing is strictly prohibited. If you already read these instructions then to get two bonus marks write the name of your favourite cloud service provider in front of the instructions heading on the question paper.
- Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.
- MCQs Answer not filled in the bubble sheet will not be considered.
- The use or possession of mobile phones and other electronic devices during the exam is strictly prohibited, whether the device is powered on or off. Do not bring mobile phones, smartwatches, tablets, or other electronic gadgets into the examination hall. If you are found with any prohibited devices, your exam may be cancelled, and further disciplinary action may be taken. Do not write below this line.

Attempt all the questions.

	Q1	Q2	Q3	Q4	Q5	Q6	Total
Total	17	12	8	8	5	45	95
Obt.	15	10	7	7	5	42.5	86.5
CLO	CLO1	CLO2	CLO3	CLO2	CLO2		

[This is the CLO 1: Describe the basics of cloud computing and related services.] Question # 1 [Avg-attempt-minutes= 20, Marks = 17]



Part A: (3 x 4 = 12 Marks)

A user named Basharat works in an AWS cloud environment and receives permissions from both an IAM user policy and an IAM group policy.

1. Create an IAM user policy in JSON that allows access to: Amazon S3, Amazon DynamoDB and explicitly denies access to Amazon Kinesis.

IAM User Policy

```

{
  "version": "2012-10-18",
  "statement": [
    {
      "sid": "AllowS3DynamoDB",
      "effect": "Allow",
      "action": ["s3.*", "dynamodb.*"],
      "resources": "*"
    },
    {
      "sid": "DenyKinesis",
      "effect": "Deny",
      "action": ["kinesis.*"],
      "resources": "*"
    }
  ]
}
  
```

2. Create another IAM group policy in JSON for a group named Developer that allows access to: Amazon S3, Amazon Ath, Amazon DynamoDB, Amazon Kinesis

IAM Group Policy

```
{
  "version": "2012-12-18",
  "statements": [
    {
      "sid": "Allow For Developer",
      "effect": "Allow",
      "action": [
        "s3.*",
        "athena.*",
        "dynamodb.*",
        "kinesis.*"
      ],
      "resource": "*",
      "conditions": {
        "stringEquals": {
          "aws:username": "Developer"
        }
      }
    }
  ]
}
```

3. Based on AWS IAM policy evaluation rules, what effective permissions (allow/deny) will Basharat have for the following services?

- Amazon S3
- Amazon DynamoDB
- Amazon Athena
- Amazon Kinesis

Effective Permissions for the User

Amazon S3 → Allowed

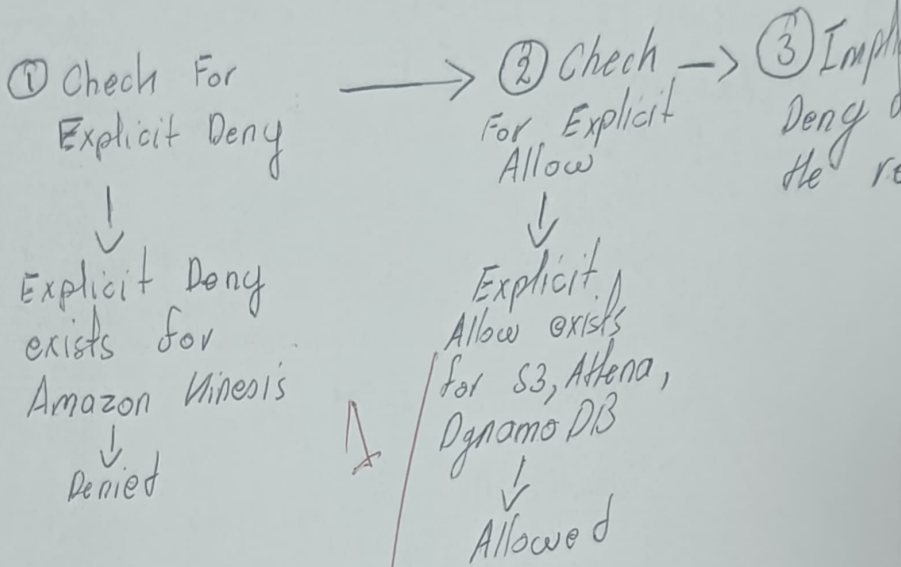
Amazon DynamoDB → Allowed

Amazon Athena → Allowed

Amazon Kinesis → ~~Allowed~~ Denied

4. Draw a permission evaluation flow diagram for this scenario.

Permission Evaluation Flow Diagram



Part B: Identify the AWS Service

(1 x 5 = 5 Marks)

Each row below describes an AWS service. Write the exact service name in the AWS Service column.

Scenario	Key Clue in Description	AWS Service
1	Discovers sensitive data in S3 using ML and pattern matching; data security risks.	AWS Macie
2	Web application firewall; monitors HTTP/HTTPS requests forwarded to protected resources.	AWS WAF
3	Analyzes root cause of security findings; uses ML, statistical analysis, graph theory; generates visualizations.	AWS Detective

continued on next page...

Scenario	Key Clue in Description	AWS Service
4	Collects security data across multiple AWS accounts and third-party products; identifies highest priority issues.	AWS Security Hub
5	Recommendations across five categories: cost optimization, security, fault tolerance, service limits, performance improvement.	AWS Trusted Advisor

[This is the CLO 2: Apply cloud computing knowledge in various real world use cases (situations).] **Question # 2** [Avg-attempt-minutes= 20, Marks = 12]

Topic: EC2 Auto Scaling

ShopFast is an e-commerce platform running its web tier on Amazon EC2 instances. Its traffic has three patterns: (i) weekend spikes that are predictable and consistent every week, (ii) flash-sale spikes that are sudden and unpredictable (up to 5× normal load), and (iii) off-peak hours where over-provisioned instances waste cost.

Part A: Elasticity, Scaling Types & Scaling Policies (12 Marks)

- Elasticity and Scaling (2 marks):** Define *elasticity* in cloud computing. How does *scaling* serve as the technique used to achieve elasticity?
- Vertical vs Horizontal Scaling (3 marks):**
 - Define vertical scaling and horizontal scaling. Give one AWS example for each.
 - ShopFast's web tier must remain available with zero downtime during scaling events. Which scaling type is more suitable and why?
- Scaling Policy Selection (4 marks):** ShopFast must choose from the following EC2 Auto Scaling policy types: *Predictive*, *Scheduled*, *Target Tracking*, *Step Scaling*. For each traffic pattern below, identify the most appropriate policy and justify your choice in one sentence.
 - Predictable, consistent traffic increase every weekend.
 - Sudden, unpredictable flash-sale traffic spikes.
 - Cost-efficient automatic scaling based on a real-time metric target (e.g. CPU utilization).
 - A historically consistent weekday/weekend pattern where ML-based forecasting can proactively scale capacity before demand arrives.
- Step Scaling (3 marks):** ShopFast configures a step scaling policy linked to a CloudWatch alarm on CPU utilization.

- What is a step adjustment and how does it differ from a single fixed scaling adjustment?
- What role does Amazon CloudWatch play in triggering step scaling?
- Give one advantage of step scaling over simple scaling.

1. It refers to the ability to adjust resources as per the demand.

Scaling allows us to increase resources (scale out) or decrease them (scale in) as per the demand metrics.

2. Vertical Scaling refers to increasing the computation power of a single resource. Eg $T2.micro \rightarrow T2.large$
Horizontal Scaling refers to increasing the number of resources to meet demand. Eg $1 * T2.micro \rightarrow 3 * T2.micro$
Horizontal Scaling is better due to no downtime, infinite increase in computation.

3. (i) $\rightarrow$ Scheduled as the traffic spike is predictable
(ii) $\rightarrow$ Step scaling as this scales proportionate to the sudden peaks.

(iii) $\rightarrow$ Target Tracking as this monitors the metrics and allows for scaling in when resources are dormant.

(iv) $\rightarrow$ Predictive as this uses the historical data + ML to determine when to scale.

4. (a) In step adjustment, the scaling occurs proportionately to the peak. ~~If the peak is above X value~~ Whereas in fixed scaling it only scales by a fixed amount of the peak.

(b) CloudWatch has alarms, which when triggered cause step scaling.

(c) Step scaling scales as per the peak so it is dynamic.

[This is the CLO 3: Analyze and architect a cloud-based solution in a real-world use case.]
Question # 3 [Avg-attempt-minutes = 15, Marks = 8]

whereas
simple
scaling
will always
scale as per a fixed amount.

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[This is the CLO 3: Analyze and architect a cloud-based solution in a real-world use case.]

Question # 3

[Avg-attempt-minutes= 15, Marks = 8]

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Topic: AWS CloudFormation Infrastructure as Code

(3 Marks)

Part A: Concepts

1. (2 marks): What is AWS CloudFormation and what problem does it solve? In your answer, explain the concept of *Infrastructure as Code* and how CloudFormation uses a template to provision resources.
2. (1 mark): What is a *CloudFormation stack*? Name two operations you can perform on a stack.

1. CloudFormation is a service provided by AWS that allows us to ~~use~~ write templates or use pre-built ones to provision infrastructure & AWS services. IaC means to define the infrastructure of cloud using code and then provision using it. CloudFormation uses template to define the architecture.

2. A collection of AWS resources created, updated or deleted as a single unit.
Operations: (1) stack create
(2) stack delete.

(5 Marks)

Part B: Template Analysis

Study the following CloudFormation template snippet and answer the questions below.

```
{
  "Resources": {
    "Ec2Instance": {
      "Type": "AWS::EC2::Instance",
      "Properties": {
        "ImageId": "ami-9d23aeea",
        "InstanceType": "m3.medium",
        "KeyName": {"Ref": "KeyPair"}
      }
    }
  },
  "Outputs": {
    "InstanceId": {
      "Description": "InstanceId",
      "Value": {"Ref": "Ec2Instance"}
    }
  }
}
```

1. (2 marks): What AWS resource type does this template create? Identify its two statically defined properties (not using Ref) and state their values.

2. (1 mark): The KeyName property uses {"Ref": "KeyPair"}. What does Ref do in a CloudFormation template?
3. (2 marks): What is the purpose of the Outputs section? What specific value does this template output, and how can you retrieve it after the stack is created?

1. → It creates an EC2 instance.
→ statically defined properties with their values
(1) "ImageId": "ami-9823aeea"
(2) "InstanceType": "m3.medium"

2. It refers to a parameter ^{or any other key} defined in the template. ^{In this it points to the key} with the name 'KeyPair'. It then uses the value given by this parameter key.

3. It outputs the values obtained during stack creation or other operations such as the an ID of an EC2 instance created.

~~We can retrieve using the getAttr~~

We can retrieve them by defining in the outputs section of the template using `Ref`.

Example: "Outputs": {
 "InstanceId": {
 "Description": "InstanceId"
 "value": { "Ref": "Ec2Instance" }
 }
}

[This is the CLO 2: Apply cloud computing knowledge in various real world use cases (situations).] Question # 4
[Avg-attempt-minutes= 15, Marks = 8]

Topic: Loose Coupling and AWS Messaging Services

DeliveryHub is a food delivery platform built on microservices. The engineering team is redesigning inter-service communication to achieve *loose coupling* and improve system resilience.

Part A: Concepts

(3 Marks)

- (1 mark): What is *loose coupling* in a distributed system? State one benefit it provides in a microservices architecture.
- (2 marks): Distinguish between *synchronous* and *asynchronous* loose coupling. For each, name **one** AWS service that achieves it (choose from: ELB · Amazon SQS · Amazon SNS · Amazon MQ).

1. Loose coupling in a distributed system refers to reducing connection surface between services. Trying to treat each service independently & reducing inter-service dependence. Benefit: Allows each component to operate independently & continue functioning even if another component fails.

2. Synchronous Loose Coupling → Reduce inter service dependence and communication through self endpoints which are synchronous. ELB.
Asynchronous Loose coupling → Decouple services. They communicate via asynchronous buffers & messages.
→ SQS, SNS, MQ

Part B: Architecture Decisions

(5 Marks)

DeliveryHub has identified the following three integration points:

#	Integration	Requirement
1	Web browser → Application servers	Must return HTTP response immediately; load must be distributed across multiple servers
2	Order service → Notification service	One order event must fan-out to Email, SMS, and Push notification consumers simultaneously
3	Order service → Inventory service	Messages must be queued; exactly one consumer processes each message

3. (3 marks): For each integration point above, recommend the most appropriate AWS service from [ELB · Amazon SQS · Amazon SNS · Amazon MQ] and justify your choice in one sentence.
4. (1 mark): What is Amazon MQ, and in what situation would you choose it over Amazon SQS?
5. (1 mark): Give one architectural advantage of using Amazon SQS between the Order Service and Inventory Service compared to a direct synchronous API call.

- 3.
- (1) → ELB. Synchronous so returns response & distributes load among servers.
- (2) → SNS as the event must be sent to multiple consumers. For out.
- (3) → SQS as there is one producer & consumer and the message must only be processed once.

4. ⇒ It is a fully managed broker provided by AWS. It is hybrid and supports both topics & queues.
- ⇒ If we are rehosting a service that uses RabbitMQ on AWS with minimal changes, Amazon MQ will be used over SQS.

5. It allows both of the services to process at their own pace without blocking each other.
- Allows us to scale either service independently.

[This is the CLO 2: Apply cloud computing knowledge in various real world use cases (situations).] Question # 5

[Avg-attempt-minutes= 10, Marks = 5]

Topic: Disaster Recovery — RPO and RTO

ArchiveSafe is a company that digitises medical records. Their system generates 200 records per hour during operating hours. The business has determined that losing more than 800 records is unacceptable. The current backup schedule runs every 6 hours. A server failure occurs at 10:00 PM; the last successful backup completed at 7:00 PM. After the failure, the IT team restores the system in 90 minutes.

- (1 mark): Calculate ArchiveSafe's *Recovery Point Objective (RPO)* in hours. Show your working.
- (1 mark): Does the current backup schedule (every 6 hours) satisfy the RPO you calculated? State the **maximum** backup interval that would satisfy the RPO.
- (1 mark): Given the backup at 7:00 PM and the failure at 10:00 PM, how many records are at risk of being lost? Does this specific incident satisfy the RPO?
- (1 mark): What is the *Recovery Time Objective (RTO)* for this incident (in hours)? At what clock time is the system fully restored?
- (1 mark): What is the **latest** time a disaster could occur after the 7:00 PM backup without violating the RPO? Justify your answer.

1. Records generated ~~by~~ per hour = 200
Number of records that can be lost = 800
$$\Rightarrow RPO = \frac{\text{No. of records that can be lost}}{\text{records generated per hour}} = \frac{800}{200} = 4 \text{ hours}$$

2. No, the current backup schedule (every 6 hrs) doesn't satisfy the RPO.
Maximum backup interval that would satisfy the RPO = 4 hours

3. No. of records lost = Records generated per hour * (Downtime - Last backup time)

$$= 200 * (10 - 7) = 600 \text{ records lost}$$

Yes this specific incident satisfy the RPO.
 Time since last backup when downtime occurred = 3 hrs < RPO 4 hrs

4. RTO is 1.5 hours (90 minutes)
 Wall Clock Time at recovery = 11:30 PM

5. 11 PM

∵ RPO is 4 hours & last backup occurred at 7 PM
 Then, the latest time a disaster can occur without violating RPO is

$$\text{Backup Point} + \text{RPO} = 7 \text{ PM} + 4 \text{ hrs} = 11 \text{ hrs}$$

∴ If system goes down at 11 PM, the max number of records lost will be 200 which is acceptable to the business.
 & downtime & time since last backup is 4 hrs ≤ RPO.

5

[This section/question is not mapped to any particular CLO.] Question # 6 [Avg-attempt-minutes = 90, Marks = 45]

Attempt All MCQs on Bubble Sheet at *LAST* Page. Return question paper with the answer sheet. Don't staple just place inside the answer sheet.

MCQ 1: A company needs to run a containerized application without provisioning or managing any EC2 servers. Which AWS service is MOST appropriate?

- a) Amazon EC2 with Auto Scaling
- b) AWS Elastic Beanstalk
- ☒ c) AWS Fargate
- d) Amazon Lightsail

MCQ 2: Which AWS compute service is BEST suited for a function that runs for 200ms, is triggered by an S3 upload event, and must scale to zero when idle?

- a) Amazon EC2
- ☒ b) AWS Lambda
- c) Amazon ECS
- d) AWS Batch

MCQ 3: An AWS service has an uptime SLA of 99.99% over a 30-day month. How many minutes of downtime are permitted?

- a) 43.2 minutes
- ☒ b) 4.32 minutes
- c) 26.3 minutes
- d) 0.432 minutes

$$30 \times 24 \times 60$$

$$43200 \times 1 \times 10^{-4}$$

$$4.32$$

MCQ 4: A video company stores 2,000 GB in Amazon S3 Standard-IA for 18 months at \$0.0125 per GB per month. What is the total storage cost?

- a) \$225.00
- b) \$337.50
- ☒ c) \$450.00
- d) \$675.00

$$1 \text{ month} = 2000 \times 0.0125$$

$$= 25$$

$$18 \text{ month} = 25 \times 18$$

MCQ 5: A startup stores 500 TB of user-generated images that must be accessible via public URL, replicated across multiple AZs, and retrieved in milliseconds. Which storage service is MOST appropriate?

- a) Amazon EBS
- b) Amazon EFS
- ☒ c) Amazon S3
- d) Amazon S3 Glacier

MCQ 6: Which statement CORRECTLY describes the key difference between Amazon EBS and Amazon EFS?

- ☒ a) EBS supports multiple EC2 instances simultaneously; EFS supports only one
- ☒ b) EFS is block storage; EBS is a file system
- c) EBS is attached to one EC2 instance at a time; EFS can be mounted by multiple EC2 instances simultaneously
- ☒ d) EFS stores objects; EBS stores files

MCQ 7: A financial company must retain audit logs for 10 years. The logs will never be accessed after 90 days but must be stored at the lowest possible cost. Which S3 storage class is MOST cost-effective?

- a) S3 Standard
- b) S3 Standard-IA
- c) S3 Glacier Flexible Retrieval
- ☒ d) S3 Glacier Deep Archive

MCQ 8: A company runs a high-performance database on an EC2 instance and needs storage with the lowest possible latency and highest consistent IOPS. Which AWS storage is MOST appropriate?

- a) Amazon S3
- b) Amazon EFS
- ☒ c) Amazon EBS Provisioned IOPS (io2)
- d) Amazon S3 Glacier

MCQ 9: The same video company (2,000 GB in S3 Standard-IA) performs 500,000 GET requests at \$0.001 per 1,000 requests over 18 months. What is the total GET request cost?

- a) \$0.05
- ☒ b) \$0.50
- c) \$5.00
- d) \$50.00

$$1 \text{ mo } \frac{500,000}{1000} \times 0.001$$

MCQ 10: Combining the previous two results: 2,000 GB stored for 18 months at \$0.0125/GB/month, plus 500,000 GET requests at \$0.001/1,000 requests. What is the TOTAL S3 cost?

- a) \$449.50
- b) \$450.00
- ☒ c) \$450.50
- d) \$451.00

MCQ 11: An RDS Multi-AZ deployment automatically fails over to a standby replica. Approximately how long does this failover process typically take?

- a) 1-5 seconds
- ☒ b) 60-120 seconds
- c) 5-10 minutes
- d) Up to 30 minutes

MCQ 12: Amazon S3 guarantees 99.99999999% (eleven 9s) annual durability. If a company stores 1 TB of data, approximately how many bytes are statistically expected to be lost per year?

- a) 0 bytes S3 has perfect durability
- ☒ b) Approximately 11 bytes
- c) Approximately 110 bytes
- d) Approximately 1,100 bytes

MCQ 13: Which of the following is the CORRECT list of all six pillars of the AWS Well-Architected Framework?

$$1 \times 10^{12} \times 10^{-11}$$

$$1 \times 10^1$$

$$1 \times 10^{12} \times 10^{-11}$$

- a) Security, Reliability, Performance, Cost, Operations, Networking
- ☒ b) Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability
- c) Scalability, Security, Reliability, Cost, Monitoring, Compliance
- d) Operational Excellence, Security, Availability, Performance, Cost Optimization, Sustainability

MCQ 14: A media company uses Amazon CloudFront for 1 year, delivering 1 TB (1,024 GB) of data at \$0.085/GB. They also make 1,000,000 GET requests annually at \$0.0075 per 10,000 requests. What is the total annual CloudFront cost?

- a) \$86.29
- b) \$87.04
- ☒ c) \$87.79
- d) \$88.54

MCQ 15: A cloud architect is designing a system where each component can operate independently and continue functioning even if another component fails. Which Well-Architected principle does this describe?

- a) Cost Optimization eliminating unused resources
- ☒ b) Reliability designing to withstand component failures using loose coupling
- c) Performance Efficiency choosing right-sized resources
- d) Operational Excellence automating operational tasks

MCQ 16: A company runs its application in a single AWS Availability Zone. According to the Well-Architected Framework, which pillar is MOST at risk?

- a) Cost Optimization
- b) Performance Efficiency
- ☒ c) Reliability
- d) Operational Excellence

MCQ 17: An Application Load Balancer (ALB) health check marks an EC2 instance as unhealthy. What happens NEXT?

- a) The instance is immediately terminated by Auto Scaling
- ☒ b) The ALB stops routing new requests to the unhealthy instance and Auto Scaling replaces it
- c) All active connections to the instance are immediately dropped
- d) The ALB continues sending traffic until a manual override is set

MCQ 18: A company wants EC2 Auto Scaling to add instances when average CPU utilization exceeds 70% and remove them when it drops below 40%. Which scaling policy type is this?

- a) Simple scaling
- b) Scheduled scaling
- c) Step scaling
- ☒ d) Target tracking scaling

MCQ 19: Which Amazon CloudWatch feature allows you to automatically take an action (such as sending an SNS notification or triggering Auto Scaling) when a metric crosses a defined threshold?

- a) CloudWatch Logs
- b) CloudWatch Metrics
- ☒ c) CloudWatch Alarms
- d) CloudWatch Dashboards

MCQ 20: An Auto Scaling group has a minimum of 2, desired of 4, and maximum of 10 instances. A scale-in event fires. What is the GUARANTEED minimum number of instances that will remain running?

- a) 0
- b) 1
- ☒ c) 2
- d) 4

MCQ 21: Which AWS service uses a push mechanism to deliver one message to multiple subscribers simultaneously?

- a) Amazon SQS
- ☒ b) Amazon SNS
- c) Amazon MQ
- d) AWS Step Functions

MCQ 22: In Amazon SQS point-to-point messaging, what is the application that places a message into the queue called?

- a) Publisher
- b) Subscriber
- ☒ c) Producer
- d) Broadcaster

MCQ 23: A company is migrating an on-premises application that uses Apache ActiveMQ to AWS. They cannot refactor the messaging code. Which AWS service is MOST appropriate?

- a) Amazon SQS
- b) Amazon SNS
- ☒ c) Amazon MQ
- d) Amazon Kinesis

MCQ 24: Which of the following is NOT a valid Amazon SNS subscriber type?

- a) AWS Lambda function
- b) Amazon Kinesis Data Firehose delivery stream
- ☒ c) Amazon RDS database instance
- d) HTTP/HTTPS endpoint

MCQ 25: Which section of an AWS CloudFormation template is REQUIRED and cannot be omitted?

- a) Parameters
- b) Outputs
- ☒ c) Resources
- d) Mappings

MCQ 26: In a CloudFormation template, {"Ref": "KeyPair"} appears in a resource property. What does this return?

- a) The ARN of the KeyPair resource
- ☒ b) The value of the KeyPair parameter or the physical ID of the KeyPair resource

- c) A hard-coded string "KeyPair"
- d) The name of the CloudFormation stack

MCQ 27: What is an AWS CloudFormation stack?

- a) A set of EC2 instances grouped in an Auto Scaling group
- b) A version-controlled repository of infrastructure templates
- ☒ c) A collection of AWS resources created, updated, or deleted as a single unit
- d) A multi-region deployment of a CloudFormation template

MCQ 28: After deploying a CloudFormation stack, a developer needs to retrieve the EC2 instance ID that was created. Where is the BEST place to expose this value?

- a) In the Resources section using a custom property
- b) In the Parameters section before deployment
- ☒ c) In the Outputs section using Ref on the resource
- d) In the Mappings section as a static value

MCQ 29: A company determines that losing up to 4 hours of data is acceptable after a failure. Which disaster recovery metric directly represents this threshold?

- a) RTO (Recovery Time Objective)
- ☒ b) RPO (Recovery Point Objective)
- c) MTTR (Mean Time to Repair)
- d) MTBF (Mean Time Between Failures)

MCQ 30: A system generates 300 records per hour. The company's RPO is 3 hours. What is the maximum number of records at risk in the worst case?

- a) 300
- b) 600
- ☒ c) 900
- d) 1,200

$$300 \times 3 = 900$$

MCQ 31: A disaster occurs at 3:00 PM. The last backup completed at 11:00 AM. The IT team fully restores the system at 5:00 PM. What is the RTO?

- a) 4 hours
- b) 6 hours
- ☒ c) 2 hours
- d) 1 hour

MCQ 32: A company's RPO is 6 hours. What is the MINIMUM backup frequency required to satisfy this RPO?

- a) Every 3 hours
- ☒ b) Every 6 hours
- c) Every 12 hours
- d) Every 24 hours

MCQ 33: An IAM user policy explicitly DENIES S3 access. The user's group policy explicitly ALLOWS S3 access. What is the effective S3 permission?

- a) Allow the group policy takes precedence over user policies
- ☒ b) Deny an explicit deny always overrides any allow
- c) Allow the most permissive policy always wins

- d) No access conflicting policies cancel each other out

MCQ 34: Which AWS service helps analyse and investigate security incidents by automatically collecting log data and using machine learning and graph theory to identify root causes?

- a) AWS Security Hub
- ☒ b) Amazon Detective
- c) Amazon Macie
- d) AWS CloudTrail

MCQ 35: A security team needs to identify which S3 buckets contain sensitive personal data (PII) using machine learning. Which service is MOST appropriate?

- a) AWS Security Hub
- b) Amazon Inspector
- ☒ c) Amazon Macie
- d) AWS Config

MCQ 36: Which AWS service provides best-practice recommendations across five categories: cost optimisation, security, fault tolerance, service limits, and performance improvement?

- a) Amazon Detective
- b) AWS CloudTrail
- ☒ c) AWS Trusted Advisor
- d) AWS Config

MCQ 37: A retailer knows its web traffic always doubles every Friday at 6:00 PM. Which EC2 Auto Scaling policy is MOST appropriate?

- a) Target Tracking Scaling
- b) Step Scaling
- ☒ c) Scheduled Scaling
- d) Predictive Scaling

MCQ 38: Which EC2 Auto Scaling policy uses historical load data and machine learning to forecast demand and pre-scale capacity BEFORE traffic arrives?

- a) Scheduled Scaling
- b) Target Tracking Scaling
- c) Step Scaling
- ☒ d) Predictive Scaling

MCQ 39: A step scaling policy adds 2 instances when CPU is 60-80%, and adds 5 instances when CPU exceeds 80%. CPU spikes to 85%. How many instances are added?

- a) 2
- ☒ b) 5
- c) 7
- d) 10

MCQ 40: An Auto Scaling group has minimum=2, desired=5, maximum=8. A scale-out event would add 4 instances. How many instances will actually be running after the event?

- a) 9
- ☒ b) 8
- c) 6
- d) 4

MCQ 41: Under the AWS Cloud Adoption Framework (CAF), which perspective is most concerned with PROCESSES, ROLES, and CULTURAL CHANGE during a migration?

- a) Business
- b) People
- ☒ c) Governance
- d) Operations

MCQ 42: Which statement about IaaS vs PaaS is CORRECT?

- a) IaaS gives you LESS control over the OS than PaaS
- b) PaaS requires the customer to patch the database engine
- ☒ c) In IaaS the customer manages the OS; in PaaS the provider manages the OS and runtime
- d) SaaS and PaaS are identical in responsibility

MCQ 43: An organisation wants the MAXIMUM upfront discount on a 3-year EC2 reservation. Which Reserved Instance payment option offers the deepest discount?

- a) No Upfront (NURI)
- b) Partial Upfront (PURI)
- ☒ c) All Upfront (AURI)
- d) On-Demand

MCQ 44: Which of the following data transfers in AWS is FREE?

- a) Outbound data from EC2 to the internet
- ☒ b) Inbound data from the internet to AWS
- c) Data transfer between Regions
- d) Data downloaded by users via CloudFront

MCQ 45: An AWS Organisations administrator attaches a Service Control Policy (SCP) that DENIES s3:CreateBucket. Which identities in a member account are affected?

- a) Only IAM users explicitly tagged
- b) Only IAM roles
- ☒ c) All identities in the account INCLUDING the root user
- d) Only programmatic access; console users are exempt

MCQ 46: Which AWS Support plan provides a dedicated Technical Account Manager (TAM)?

- a) Basic
- b) Developer
- c) Business
- ☒ d) Enterprise

MCQ 47: Which AWS service appears as a SINGLE GLOBAL resource view in the console (NOT per-Region)?

- a) Amazon EC2
- b) Amazon S3
- c) AWS Lambda
- ☒ d) AWS IAM

MCQ 48: Which statement about AWS Availability Zones (AZs) is TRUE?

- a) AZs within a Region are typically 100 km or more apart for disaster isolation
- b) AZs are connected via the public internet
- ☒ c) AZs are physically isolated facilities within a Region, connected by high-bandwidth low-latency private links
- d) Each AZ is a single physical data centre

MCQ 49: An Edge Location is BEST described as:

- a) A miniature Region used for compute workloads
- ☒ b) A CloudFront caching node used to deliver content closer to end users
- c) A failover AZ activated during outages
- d) A data centre dedicated to AWS Lambda

MCQ 50: Under the AWS Shared Responsibility Model, who patches the GUEST OPERATING SYSTEM running on an Amazon EC2 instance?

- a) AWS
- ☒ b) The customer
- c) Shared 50/50
- d) The third-party AMI publisher

MCQ 51: An IAM user has NO inline or attached policies. They attempt to read an S3 bucket whose bucket policy explicitly grants s3:GetObject to their ARN. Can they read the object?

- a) No — they have no IAM policy granting access
- ☒ b) Yes — the bucket policy explicitly allows them and nothing explicitly denies
- c) No — both identity-based AND resource-based allows are always required
- d) Only if the user is also in an IAM group

MCQ 52: An IAM user has an attached policy that ALLOWS ec2:*. A separate Service Control Policy (SCP) on the AWS Organisation DENIES ec2:TerminateInstances. Can the user terminate an EC2 instance?

- a) Yes — IAM allow overrides organisational policy
- ☒ b) No — SCP deny intersects with the IAM allow and blocks the action
- c) Yes, only via CLI but not console
- d) Yes, but the action is reverted automatically

MCQ 53: What permissions does a NEWLY CREATED IAM user have by default (with no policies attached)?

- a) Read-only access across all services
- b) Full administrator access until policies are explicitly attached
- ☒ c) None — all actions are implicitly denied
- d) Access only to billing dashboard

MCQ 54: Which of the following is NOT a valid second factor for AWS Multi-Factor Authentication (MFA)?

- a) Virtual MFA app (Google Authenticator, Authy)
- b) U2F security key
- c) Hardware MFA token (key fob or display card)
- ☒ d) The user's IAM password re-entered

MCQ 55: Which IAM entity is BEST suited for granting AWS permissions to an EC2 instance so that an application running on it can call AWS APIs?

- a) An IAM user with access keys baked into the AMI
- ☒ b) An IAM role attached to the instance via an instance profile
- c) The root user credentials stored in an environment variable
- d) A bucket policy attached to the instance

MCQ 56: A subnet's Network ACL has rule 100 (ALLOW HTTP from 0.0.0.0/0) and rule 50 (DENY HTTP from 203.0.113.7). HTTP traffic arrives from 203.0.113.7. What happens?

- ☒ a) Allowed — rule 100 matches all sources
- b) Denied — NACL rules evaluate lowest-number first, so rule 50 fires
- c) Denied — NACLs default to deny on any conflict
- d) Allowed — DENY rules are not supported in NACLs

MCQ 57: Which statement correctly contrasts a Security Group with a Network ACL (NACL)?

- ☒ a) Both are stateless and operate at the subnet level
- b) Security Group is stateful and operates at the instance level; NACL is stateless and operates at the subnet level
- c) Security Group is stateless; NACL is stateful
- d) Both are stateful but NACL allows only inbound rules

MCQ 58: What component is REQUIRED in a public subnet so that EC2 instances in a PRIVATE subnet can download OS patches from the public internet?

- a) An Internet Gateway placed in the private subnet
- ☒ b) A NAT Gateway placed in the PUBLIC subnet, with a route from the private subnet to the NAT
- c) A NAT Gateway placed in the private subnet
- d) A VPN Connection

MCQ 59: A web app's users are spread across the US, Europe, and Asia. Which Route 53 routing policy directs each user to the AWS Region that gives them the LOWEST observed latency?

- a) Geolocation routing
- b) Weighted routing
- ☒ c) Latency-based routing
- d) Failover routing

MCQ 60: An EC2 instance is in a private subnet with no route to the internet. The instance can still access Amazon S3 in the same Region WITHOUT a NAT Gateway. Which AWS feature makes this possible?

- ☒ a) VPC Endpoint (Gateway type for S3)
- b) Internet Gateway with private routing
- c) Elastic IP attached to the instance
- d) Direct Connect

MCQ 61: AWS Lambda imposes a HARD maximum runtime per invocation. What is it?

- a) 60 seconds
- b) 5 minutes
- ☒ c) 15 minutes

d) 1 hour

MCQ 62: An EC2 instance uses INSTANCE STORE volumes (NOT EBS) for fast scratch storage. The developer issues "Stop" from the console. What happens to data on the instance store?

- a) Preserved — instance store survives stop/start
- ☒ b) Permanently lost — instance store is ephemeral and discarded on stop
- c) Automatically migrated to EBS
- d) Automatically migrated to S3 Glacier

MCQ 63: Which EC2 pricing model can offer up to 90% discount but can be reclaimed by AWS with approximately 2 minutes of notice?

- a) On-Demand
- b) Reserved Instances (3-year AURI)
- ☒ c) Spot Instances
- d) Savings Plans

MCQ 64: An AMI (Amazon Machine Image) is BEST described as containing:

- a) Only the operating system
- ☒ b) A template with OS, application server, applications, launch permissions, and block-device mapping
- c) Only the data volumes attached to an EC2 instance
- d) A snapshot of running RAM and CPU registers

MCQ 65: Amazon S3 promises 11 nines (99.999999999%) DURABILITY. What does durability measure?

- a) Uptime — how often the service is reachable
- ☒ b) The probability of NOT losing an object in a given year
- c) Replication speed between Regions
- d) The percentage of requests served from the closest edge

MCQ 66: Which statement about Amazon EBS is TRUE?

- a) An EBS volume can be attached to any EC2 instance in any AZ within the Region
- ☒ b) An EBS volume can be attached to only ONE EC2 instance at a time and must be in the SAME AZ as the instance
- c) EBS volumes are mounted via NFS like EFS
- d) EBS volumes survive only while the instance is running

MCQ 67: Which S3 storage class has the LOWEST cost per GB but a MINIMUM retrieval time of 12 hours?

- a) S3 Standard-IA
- b) S3 One Zone-IA
- c) S3 Glacier Flexible Retrieval
- ☒ d) S3 Glacier Deep Archive

MCQ 68: S3 versioning is enabled on a bucket. A user deletes an object. What is the OBSERVABLE outcome?

- a) The object is permanently and immediately erased
- ☒ b) A delete-marker is placed at the top; the prior version can be restored by removing the marker
- c) The object moves to S3 Glacier automatically

d) The object is moved to a "Trash" prefix for 30 days

MCQ 69: Standard-IA vs One Zone-IA — which statement is TRUE?

- a) Both store data across 3+ AZs
- ☒ b) Standard-IA stores across ≥ 3 AZs; One Zone-IA stores in a SINGLE AZ at ~20% lower cost
- c) One Zone-IA has higher durability than Standard-IA
- d) Standard-IA has retrieval fees but One Zone-IA does not

MCQ 70: An RDS database requires SYNCHRONOUS replication to a standby in a different AZ with AUTOMATIC failover and zero data loss. Which feature provides this?

- a) Read Replicas
- ☒ b) Multi-AZ Deployment
- c) Manual DB snapshots
- d) Cross-Region replication

MCQ 71: Which AWS database service is BEST suited for petabyte-scale OLAP analytical queries on columnar data?

- a) Amazon RDS for MySQL
- b) Amazon DynamoDB
- c) Amazon Aurora
- ☒ d) Amazon Redshift

MCQ 72: In a DynamoDB table with a COMPOSITE primary key, what makes up the composite?

- a) Two partition keys
- b) Partition key (hash) + Sort key (range)
- ☒ c) Partition key + Secondary index key
- d) Username + Account ID

MCQ 73: Which read mode in DynamoDB returns data that MAY reflect a stale value but at LOWER cost and higher throughput?

- a) Strongly consistent reads
- ☒ b) Eventually consistent reads
- c) ACID-isolated reads
- d) Transactional reads

MCQ 74: Amazon Aurora claims to be HOW MANY times faster than standard MySQL on the same hardware?

- a) 1.5x
- b) 3x
- ☒ c) 5x
- d) 10x

MCQ 75: Which Well-Architected design principle MOST directly supports the idea that "components can operate independently and continue functioning if other components fail"?

- a) Stop guessing capacity
- ☒ b) Loose coupling
- c) Automate with code
- d) Implement strong identity foundation

MCQ 76: An organisation says: "Our RTO is 4 hours and our RPO is 15 minutes." This means:

- a) 4 hours of downtime tolerated, 4 hours of data loss tolerated
- b) 15 minutes of downtime tolerated, 15 minutes of data loss tolerated
- ☒ c) Up to 4 hours of downtime AND up to 15 minutes of data loss tolerated
- d) Up to 15 minutes of downtime AND up to 4 hours of data loss tolerated

MCQ 77: Which AWS Trusted Advisor category does NOT exist?

- ☒ a) Cost Optimisation
- b) Security
- c) Fault Tolerance
- ☒ d) Application Refactoring

MCQ 78: Which AWS load balancer is BEST suited for handling MILLIONS of TCP requests per second with ultra-low latency for a gaming workload?

- a) Application Load Balancer (ALB)
- ☒ b) Network Load Balancer (NLB)
- c) Classic Load Balancer (CLB)
- d) Gateway Load Balancer (GWLB)

MCQ 79: What is the DEFAULT cooldown period for an EC2 Auto Scaling group?

- a) 30 seconds
- b) 60 seconds
- ☒ c) 300 seconds (5 minutes)
- d) 600 seconds (10 minutes)

MCQ 80: A CloudWatch metric recorded at 1-MINUTE granularity is retained for how long by default?

- ☒ a) 1 day
- b) 15 days
- c) 63 days
- d) 455 days

MCQ 81: An organisation is migrating 100 TB of data to AWS but has only a 100 Mbps internet uplink. Which is the FASTEST realistic option?

- a) Upload directly over the internet
- b) Use AWS DataSync over the existing internet link
- ☒ c) Order an AWS Snowball device, copy locally, and ship it back
- d) Open a Direct Connect circuit and upload over it

MCQ 82: A company stores 4,000 GB in S3 Standard at \$0.023 per GB per month for 12 months. What is the total storage cost?

- a) \$552.00
- b) \$920.00
- ☒ c) \$1,104.00
- d) \$2,208.00

MCQ 83: A company stores 3,000 GB in S3 Standard-IA at \$0.0125 per GB per month for 24 months. What is the total storage cost?

- a) \$375.00
- b) \$750.00
- ☒ c) \$900.00

d) \$1,800.00

MCQ 84: An application performs 2,000,000 GET requests on S3 Standard at \$0.0004 per 1,000 requests. What is the total GET request cost?

a) \$0.40

☒ b) \$0.80

c) \$8.00

d) \$80.00

MCQ 85: An application performs 1,000,000 PUT requests on S3 Standard at \$0.005 per 1,000 requests. What is the total PUT cost?

a) \$0.50

☒ b) \$5.00

c) \$50.00

d) \$500.00

MCQ 86: Amazon CloudFront delivers 2 TB (use 1 TB = 1,024 GB) of data at \$0.085 per GB. What is the data transfer cost?

a) \$87.04

☒ b) \$174.08

c) \$348.16

d) \$2,048.00

MCQ 87: Amazon CloudFront serves 5,000,000 GET requests at \$0.0075 per 10,000 requests. What is the request cost?

a) \$0.375

☒ b) \$3.75

c) \$37.50

d) \$375.00

MCQ 88: Combine the previous two CloudFront figures: 4 TB of data at \$0.085/GB and 4,000,000 GET requests at \$0.0075 per 10,000. What is the TOTAL CloudFront cost?

a) \$174.83

b) \$348.16

☒ c) \$351.16

d) \$355.66

348.16 + 3

MCQ 89: Amazon S3 advertises 11 nines (99.999999999%) annual durability. If a customer stores 5 TB, approximately how many bytes are statistically expected to be lost per year? (Use 1 TB = 2^{40} bytes.)

a) ~ 11 bytes

b) ~ 33 bytes

☒ c) ~ 55 bytes

d) ~ 110 bytes

$5 \times 2^{40} \times 10^{-11}$

MCQ 90: An object is uploaded to S3 and accessed multiple times per day for the first 30 days. From day 31 to day 180 it is accessed only once a quarter. Which S3 lifecycle TRANSITION at day 31 minimises cost while preserving millisecond-latency access?

a) S3 Standard → S3 Glacier Deep Archive

b) S3 Standard → S3 Standard-IA

☒ c) S3 Standard → S3 Glacier Flexible Retrieval

d) Keep on S3 Standard — no transition is cheaper

$5 \times 10^{12} \times 10^{-11}$

— MCQ Bubble Sheet —

Name: <u>Muhammad Zubair Adnan</u>
Exam: <u>CS-4075</u> Date: <u>11-6-2026</u>

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— GOOD LUCK —